SAFEHER: WOMEN-ONLY CAB AND PERSONAL SAFETY APPLICATION

A REPORT

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1. INTRODUCTION

1.1 General

SafeHer is an Android application concept focused on safer travel and emergency assistance for women. The application interface shown in the project demonstration combines women-oriented cab booking information with personal-safety tools. Its main screen presents a journey-planning flow, ride categories and driver information, while a separate emergency screen provides an SOS control, deterrent tools and emergency call shortcuts.

The project is developed in Android Studio using Java. It brings travel-related information and quick-access safety actions together in a mobile interface. The design uses a pink and white visual theme, clear action buttons and bottom navigation to make the main sections easy to reach.

1.2 Problem Statement

People travelling alone may want additional reassurance before and during a trip, as well as quick access to trusted contacts or emergency services. Switching between a cab service, phone dialler and safety information can add friction at a stressful moment. SafeHer explores a single application interface that combines journey planning, ride and driver details, emergency actions, safety guidance and trip/incident history.

1.3 Objectives

- Design an Android interface for planning a journey and viewing women-oriented cab options.
- Display ride categories, fare information and driver verification/rating details.
- Provide a prominent SOS action intended to share location with guardians and raise an alarm.
- Offer quick access to emergency helplines and deterrent/evidence tools.
- Present cab-safety guidance, transport-protection information and recent trip or incident logs.
- Demonstrate the use of Android Studio and Java to build a mobile application interface.

1.4 Scope of the Project

The demonstrated scope includes the user interface and flows visible in the emulator recording: safe-journey planning, ride selection, emergency SOS, emergency call shortcuts, safety advice and trip/incident logs. The recording does not independently establish whether every action is connected to live cab dispatch, production GPS tracking, cloud storage, or real emergency

services. These capabilities should therefore be treated as intended functionality unless they are implemented and verified separately.

2. SYSTEM OVERVIEW AND TECHNOLOGIES

2.1 Application Overview

SafeHer is organised around a small set of task-focused screens accessible through bottom navigation. The journey screen contains pickup and drop-off fields, suggested destinations, ride-category cards, fare labels and driver information. The emergency screen centres on a large SOS button and includes options such as police siren, audio recording and direct dialling of emergency numbers. Additional screens present safety tips, legal/transport information and a record of recent trips and incidents.

2.2 Development Tools and Technology

- Android Studio: Integrated development environment used to build and run the Android application.
- Java: Programming language used for application logic and event handling.
- Android Emulator: Virtual Android device used to demonstrate the screens and interactions.
- Android UI components: Layouts, text labels, buttons, cards and navigation elements used to organise the interface.
- SQLite (indicated by the trip/incident log screen): The interface labels the recent trip and incident log as SQLite; the exact database schema and persistence behaviour should be documented from the source code.

2.3 Main Modules

A. Safe Journey Planning and Ride Selection

This screen lets the user review a pickup location and destination, choose from suggested locations, and compare ride categories. The visible interface shows SafeHer Sedan and SafeHer Mini options with displayed fares. A driver card presents a driver name, vehicle details, rating and a verification label before the user requests a ride.

B. Emergency SOS

The SOS screen presents a large, central action and explains that it can dispatch GPS coordinates to guardians and sound an alarm. The screen also displays a shake-to-SOS instruction, indicating a proposed shortcut for triggering emergency assistance.

C. Emergency Deterrent and Evidence Tools

Two prominent controls are labelled Police Siren and Record Audio. The interface also indicates the audio evidence recorder status. Actual microphone permissions, recording behaviour, storage, and background restrictions depend on the implementation and must be tested on a physical device.

D. Emergency Speed Dial

The screen includes shortcuts labelled 112 (national emergency), 1091 (women), 100 (police) and 108 (ambulance). These controls are intended to reduce the steps required to contact assistance. Availability and dialling behaviour should be confirmed on the target device and network.

E. Safety Guidance and Trip/Incident Log

A guidance screen lists five cab-safety rules, including verifying the vehicle and driver, checking child locks, sharing the trip PIN only when appropriate, using the SOS shortcut and keeping live route sharing active. It also presents transport-protection information and a recent trip/incident log with entries such as SOS dispatched and trip completed.

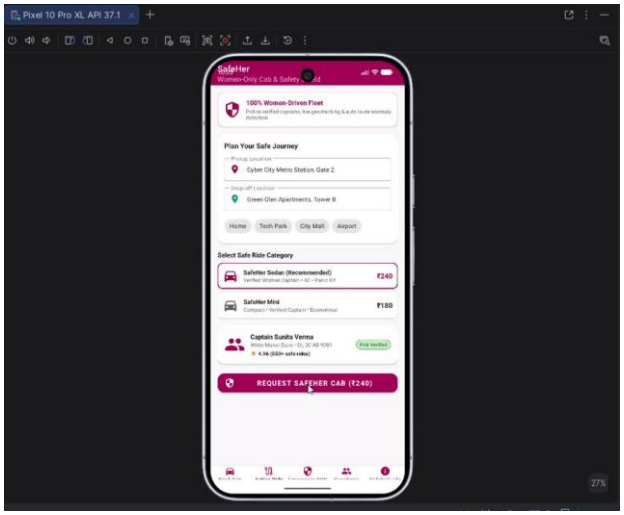
3. DESIGN AND IMPLEMENTATION

3.1 User Interface Design

The interface uses a consistent magenta/pink, white and teal palette. Cards separate related information, while filled buttons emphasise important actions. The emergency screen uses a red SOS control to create a clear visual focus. The bottom navigation provides persistent access to the major application areas. The design is optimised for a portrait phone display as shown in the Android emulator.

3.2 Safe Journey Planning

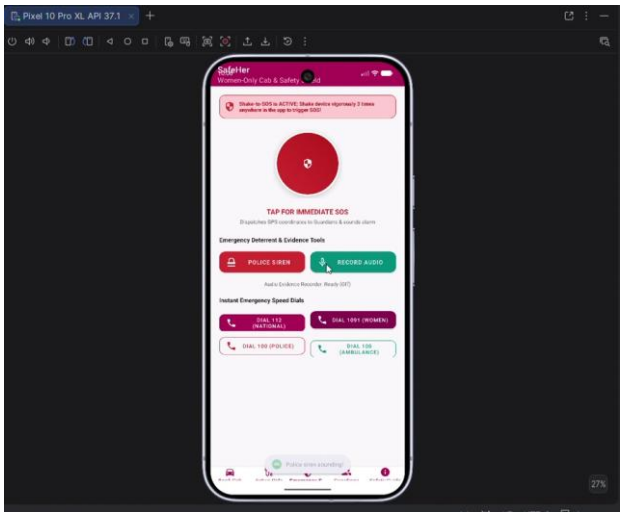
The journey-planning screen groups related tasks in a top-to-bottom order: a women-driven fleet information banner, pickup and drop-off fields, suggested destinations, ride category selection, driver information and a request button. Displayed example fares in the demonstration are ₹240 for SafeHer Sedan and ₹180 for SafeHer Mini. These values are demonstration data and should not be interpreted as live prices.



Safe journey planning and ride selection screen.

3.3 Emergency and SOS Features

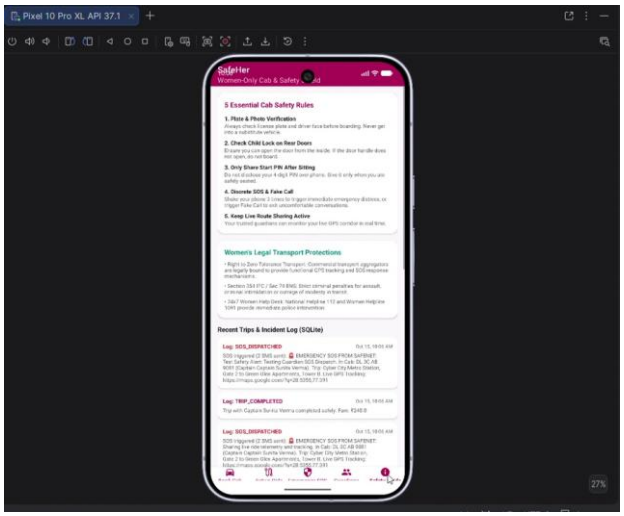
The emergency screen is designed to make urgent actions easy to identify. A prominent SOS button is accompanied by explanatory text. Separate buttons provide the siren and audio-recording options, and four speed-dial controls are visible beneath them. Since emergency functionality can affect user safety, the implementation should request permissions clearly, provide confirmation where appropriate, handle failures gracefully and avoid implying that help has been contacted unless the action succeeds.



Emergency SOS, deterrent tools and emergency dial shortcuts.

3.4 Safety Information and Trip Logs

The safety information screen provides a checklist of practical cab-safety guidance and a section on legal transport protections. Below it, the recent trip and incident log displays entries with status labels and trip details. This screen helps users review recent activity and keeps safety guidance available within the same application.



Cab-safety guidance and recent trip/incident history.

4. TESTING AND OBSERVATIONS

The following observations are based on the screen-recording demonstration. They describe visible interface behaviour and are not a substitute for functional testing of the source code on a device.

TEST AREA	OBSERVATION	STATUS
Launch and navigation	Application opens to the SafeHer interface; bottom navigation is visible.	Observed in demonstration
Journey screen	Pickup/drop-off details, destination shortcuts, ride cards, fares and driver details are displayed.	Observed in demonstration
Ride request	A request button is present for the selected cab option.	Button visible; backend dispatch not verified
Emergency screen	SOS, Police Siren, Record Audio and emergency dial controls are displayed.	Observed in demonstration
SOS response	The interface describes GPS sharing and alarm behaviour; a shake-to-SOS instruction is shown.	End-to-end delivery not independently verified
Safety information	Cab safety rules and transport-protection information are displayed.	Observed in demonstration
Trip/incident log	Recent entries with SOS-dispatched and trip-completed statuses are shown.	Screen shown; persistence not independently verified

Recommended additional tests include checking Android runtime permissions for location, microphone and phone calls; validating input fields; testing with location services disabled; checking SOS failure and cancellation paths; verifying whether logs persist after restarting the app; and confirming accessibility through readable text sizes, contrast and screen-reader labels.

5. CONCLUSION AND FUTURE SCOPE

SafeHer demonstrates a mobile application concept that combines women-oriented cab journey planning with emergency and personal-safety features. The interface brings together ride categories and driver details, an SOS screen, emergency call shortcuts, deterrent/evidence controls, safety guidance and trip/incident history. The project illustrates how Android Studio and Java can be used to organise multiple safety-related functions into a single Android experience.

Future development can focus on connecting the interface to verified backend services, implementing reliable live-location sharing with explicit consent, adding trusted-contact management, securing trip and incident data, and testing emergency actions across Android versions and devices. Further improvements may include accessibility testing, privacy notices, secure authentication, automated unit/UI tests, and clear feedback when a requested action

succeeds or fails. Any deployment involving emergency assistance should be thoroughly tested and should not promise protection that the system cannot reliably provide.

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5. Project report format supplied by the user: “report format NMIMS.docx”.